

INDIVIDUAL CASE SAFETY REPORT (ICSR)

PROCESSING: CASE 3

Section 1: Literature Intake Sheet

Field	Value
Article ID	Cureus 16(7): e64081
DOI	https://doi.org/10.7759/cureus.64081
Journal	Cureus
Year	2024 (July)
Author	Kristina L. Quan Soon, Sundeep Shah, Ramy Ibrahim, Khalid Alzwareh
Country	United States
Patient Age	54 years old
Sex	Female
Drug	Methotrexate
Reaction	Iatrogenic Methotrexate Toxicity (Pancytopenia, Oral Mucositis, Colitis, Dermatitis)
Serious?	Yes
Reporter Type	Healthcare Professional (Physician)
Source	Literature Case Report

Section 2: Case Triage

- **Patient?** Yes (54-year-old female)
- **Reporter?** Yes (Primary Care Physician / Hospitalists)
- **Drug?** Yes (Methotrexate)
- **Drug reaction?** Yes (Pancytopenia, mucositis, colitis, skin lesions)

Triage Decision: VALID ICSR. All four minimum criteria for an Individual Case Safety Report are present.

Section 3: Data Extraction Sheet

Parameter	Value
Patient ID	Case003
Age	54 years
Gender	Female
Weight	Not reported
History	Mixed connective tissue disease, Hypertension, Lyme disease, Sick sinus syndrome, Ulcerative colitis
Medical History	Polyarthritis secondary to mixed connective tissue disease
Suspect Drug	Methotrexate (2.5 mg tablets)
Dose	7.5 mg (3 tablets) twice a day (Total: 15 mg daily)
Frequency	Twice daily for 5 consecutive days (Medication administration error; intended dose was 15 mg once weekly)
Route	Oral
Lot Number	Not reported

Start Date	Unspecified (administered for 5 consecutive days prior to admission)
Stop Date	Discontinued after 5 days upon identification of error
Concomitant Drugs	Mesalamine, Budesonide (Both stopped during admission)
Lab Values	Admission (Day 1): WBC: 0.7 K/uL, Hb: 11 g/dL, Platelets: 133 K/uL, Neutrophils: 12%, ANC: 0.09 K/uL, K+: 3.1 mmol/L, Creatinine: 0.59 mg/dL. Nadir (Day 4/5/6): WBC: 0.5 K/uL, Platelets: 15 K/uL, ANC: 0.01 K/uL. Serum Methotrexate: <0.04 mg/ml.
Reaction	Severe iatrogenic methotrexate toxicity presenting as pancytopenia (severe leukopenia, thrombocytopenia, anemic worsening), oral hemorrhagic sores, dysphagia, nausea, vomiting, diarrhea, hematochezia, inflammatory mucositis, colitis, and erythematous papules.
Outcome	Recovered
Action Taken	Drug permanently withdrawn; Leucovorin rescue initiated
Dechallenge	Positive (Hematological indices and mucosal lesions gradually normalized following drug withdrawal and antidote administration)
Rechallenge	Not performed
Reporter Assessment	Accidental iatrogenic overdose leading to acute multi-systemic methotrexate toxicity.

Section 4: Narrative Writing

Clinical Narrative

Case Summary: A 54-year-old female patient with a medical history of mixed connective tissue disease, ulcerative colitis, hypertension, Lyme disease, and sick sinus syndrome experienced

severe, life-threatening multi-systemic toxicity following an accidental iatrogenic methotrexate overdose.

Onset and Medication Error: The patient was prescribed oral methotrexate 15 mg once weekly for polyarthritis. Due to a medication administration error, she erroneously ingested three 2.5 mg tablets twice daily (15 mg daily) for 5 consecutive days. She initially presented to the emergency room (ER) with general malaise and was erroneously discharged with antibiotics for suspected folliculitis. The following day, she presented to her primary care physician with worsening malaise, icterus, and significant dysphagia. Upon confirmation of the overdose, she was urgently referred to a tertiary care center.

Clinical Progression and Hospitalization: Upon admission, the patient was febrile (38.5°C) and exhibited severe oral hemorrhagic sores on the hard and soft palate, significant dysphagia, aphonia, nausea, and vomiting. Skin examination revealed erythematous papules on the buttocks and posterior lower limbs. During hospitalization, she developed severe diarrhea, hematochezia, extreme fatigue, generalized weakness, and worsening skin lesions. A nasogastric tube was placed due to severe oral mucositis. An abdominal CT scan confirmed acute inflammatory mucositis and colitis.

Laboratory and Diagnostic Investigations: Serial blood work revealed profound myelosuppression/pancytopenia. White blood cell count reached a nadir of 0.5 K/uL, platelets dropped to 15 K/uL, and the absolute neutrophil count (ANC) fell to a critical nadir of 0.01 K/uL. Her basic metabolic profile revealed initial hypokalemia (3.1 mmol/L), while renal function parameters remained normal. Serum methotrexate levels were found to be <0.04 mg/ml.

Management and Outcome: Methotrexate, mesalamine, and budesonide were stopped immediately. Multidisciplinary care was coordinated with gastroenterology, hematology, and wound care. Treatment included intravenous leucovorin rescue, filgrastim (G-CSF), folic acid, intravenous broad-spectrum antibiotics, anti-emetics, acetaminophen, anticoagulants, and topical antifungals. After 6 days of stabilization, she was transferred to a general hospital for 7 additional days of IV antibiotic therapy and monitoring. Following a total of 13 days of hospital management, her hematological parameters fully normalized (WBC: 7.5 K/uL, Platelets: 299 K/uL, ANC: 4.4 K/uL). She completed 14 days of intensive nursing at a skilled nursing facility, achieving complete resolution of weakness and mucositis.

Section 5: MedDRA Coding

MedDRA Hierarchy	Terminology
Lowest Level Term (LLT)	Methotrexate toxicity
Preferred Term (PT)	Drug toxicity
High Level Term (HLT)	Poisoning and toxicity

High Level Group Term (HLGT)	Overdoses and underdosing NEC
System Organ Class (SOC)	Injury, poisoning and procedural complications

Secondary clinical events include: Pancytopenia (SOC: Blood and lymphatic system disorders), Stomatitis hemorrhagic (SOC: Gastrointestinal disorders), Colitis (SOC: Gastrointestinal disorders).

*****MedDRA numeric codes omitted because the terminology is licensed; hierarchy shown for educational purposes***

Section 6: Seriousness

Seriousness: Yes

Reason for Seriousness:

- ☐ Death
- ☒ Life-threatening (Severe febrile neutropenia with ANC of 0.01 K/uL)
- ☒ Hospitalisation (Required 13 days of acute inpatient care)
- ☐ Congenital anomaly
- ☐ Disability
- ☒ Other medically important condition (Severe multi-systemic overdose toxicity)

Section 7: Expectedness

Classification: Expected

Evidence: According to the standard reference safety information (USPI and SmPC) for Methotrexate, multi-systemic toxicities including profound myelosuppression (pancytopenia, leukopenia, thrombocytopenia), severe gastrointestinal ulceration, mucositis, colitis, and intestinal hemorrhage are well-documented, dose-dependent adverse reactions. Labeling contains black box warnings highlighting that accidental daily administration instead of weekly dosing is a known source of fatal iatrogenic toxicity, requiring immediate drug cessation and leucovorin rescue therapy.

Section 8: Causality Assessment

WHO-UMC Causality Category

Category: Probable / Likely

Reason: The severe toxic multi-organ syndrome followed a definitive chronological timeline directly after a verified 5-day high-dose medication overdose, responded predictably to drug

withdrawal combined with an official antagonist (leucovorin), and cannot be reasonably attributed to the patient's stable underlying diseases.

Naranjo Score Assessment

Question	Score
1. Are there previous conclusive reports on this reaction?	Yes (+1)
2. Did the adverse event occur after the suspected drug was administered?	Yes (+2)
3. Did the adverse reaction improve when the drug was discontinued or a specific antagonist was administered?	Yes (+1)
4. Did the adverse reaction reappear when the drug was readministered?	No / Not Done (0)
5. Are there alternative causes that could have on their own caused the reaction?	No (+2)
6. Did the reaction reappear when a placebo was given?	No / Not Done (0)
7. Was the drug detected in the blood or other fluids in concentrations known to be toxic?	No (0) (Note: Serum levels were <0.04 mg/ml, which is expected in delayed/chronic tissue toxicity)
8. Was the reaction more severe when the dose was increased or less severe when decreased?	Yes (+1) (Accidental daily dosing instead of weekly directly caused severity)
9. Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	No / Unknown (0)
10. Was the adverse event confirmed by any objective evidence?	Yes (+1) (Confirmed via objective serial CBC nadirs and

	CT scan confirming colitis)
Total Naranjo Score	+8

Causality Interpretation: Probable (A score of 5–8 indicates a probable relationship between the medication error/overdose and the subsequent adverse clinical events).

Section 9: Quality Check

Check Item	Status	Comments
Missing patient info	Pass	Age, gender, medical history fully extracted. Weight missing from literature.
Dates logical	Pass	Time course from onset, nadir, and discharge processing is logical.
Duplicate events	Pass	No duplicate case processing files noted.
Narrative complete	Pass	Reviewed and validated against source text layout. (Reviewer initials: MJ, 19 July, 2026)
MedDRA verified	Pass	Terms coded accurately using standard hierarchy layout.
Seriousness correct	Pass	Criteria for life-threatening complications and hospitalization met.
Causality complete	Pass	WHO-UMC and Naranjo assessments match criteria guidelines.
CIOMS Form	Complete	Logged and formatted for output generation.